

Bailey E. Quinn

baileyequinn@gmail.com | 352.213.2198
https://darkghastful.github.io/Portfolio

PUBLICATIONS

- [1] B. E. Quinn, J. A. R. Rodríguez, E. K. Sam, et al. The nasal microbiome in early infancy is primarily shaped by the maternal nasal microbiome. *Journal of Allergy and Clinical Immunology*, Volume 156, Issue 3, 668 - 678.

EXPERIENCE

WASHINGTON UNIVERSITY IN ST. LOUIS MEDICAL SCHOOL DIVISION OF INFECTIOUS DISEASE | BIOINFORMATICIAN

Sept 2024 - Aug 2025 | St. Louis, MO

- Built reusable ETL pipelines in R/Python utilizing REST APIs and headless Chrome to ingest, clean, and analyze 0.5–2 TB datasets; parameterized for reproducibility.
- Designed an R package to map DNA mutations to mutated protein sequences and premature truncation using live API queries.
- Administrated lab databases and sequence stores for data validation, summarization, and analysis.
- Facilitated large-scale data transfers (3–9 TB) with verification.

WASHINGTON UNIVERSITY IN ST. LOUIS MEDICAL SCHOOL DIVISION OF ALLERGY AND IMMUNOLOGY | BIOINFORMATICIAN

Aug 2022 - Sept 2024 | St. Louis, MO

- Conducted analytics for a three-hospital cohort study; built ETL pipelines in R, executed PERMANOVA/ordination/clustering, and was first author on a the corresponding publication.
- Collaborated with both clinicians and researchers to translate findings into clear visuals and summaries for publication and presentations.
- Administrated lab databases and sequence stores for data validation, summarization, and analysis

UNIVERSITY OF MISSOURI | HONORS FELLOW / UNDERGRADUATE RESEARCH ASSISTANT

Aug 2018 - May 2021 | Columbia, MO

- Created a lesson plan for a classroom accessible experiment implemented in Missouri middle schools through a science in education program.
- Maintained a line of macrophage cells and assisted in experiments.

ORGANIZATIONS

SOUTH BROADWAY ART PROJECT

| WORKING MEMBER

Mar 2024 – Present | St. Louis, MO

MIZZOU STEEL BRIDGE TEAM

| GRAPHICS

Aug 2018 – May 2022 | Columbia, MO

BIOCHEMISTRY CLUB

Aug 2018 – May 2022 | Columbia, MO

COLLEGE MENTOR FOR KIDS

Aug 2019 – Dec 2021 | Columbia, MO

EDUCATION

UNIVERSITY OF MISSOURI

BACHELOR OF SCIENCE IN

BIOCHEMISTRY

2022 | Columbia, MO

GPA 3.4

SKILLS

LANGUAGES

Python • R • SQL • Linux • Bash

ANALYTICS AND ML

regression • conditional diffusion • experimental design

DATA ENGINEERING

REST API integration • headless scraping • version control

PLATFORMS

Jupyter • Linux • Docker • Git • Markdown

PRESENTATIONS

DOCKER/18S WORKSHOP

Nov 2023 | St. Louis, MO

Created and ran an interactive workshop designed to teach the processing and storage of sequencing data, an introduction to statistical analytics, and how to use docker images on a remote server.

REGENERATIVE MEDICINE WIP

Apr 2023 | St. Louis, MO

Presented my research detailing the transmission of bacteria from the maternal nasal microbiome to the infant nasal microbime.

ACHIEVEMENTS

BLACK AND GOLD/ALUMNI SCHOLAR

Aug 2018 - May 2022 | Columbia, MO

HONORS DISCOVERY FELLOWSHIP

Aug 2018 - May 2019 | Columbia, MO